Bangladesh Army University of Engineering & Technology

Class test: 01, Course Code: **MATH-1241**,

Course title: Ordinary, Partial Differential Equations and Coordinate Geometry

Time: **35** mintues

1. What do you understand by order and degree of a differential equation? Write down a third order and second degree ordinary differential equation. 4
2. Assess the differential equations of all circles in the XOY which have their centers on the x axis and have given radius. 5
3. Explain homogeneous differential equation. Identify and solve the differential equation $x^2 y\,dx - \left(x^3 + y^3\right) dy = 0$. 6

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Class test: 02, Course Code: **MATH-1241**,

Course title: Ordinary, Partial Differential Equations and Coordinate Geometry

1. What do you mean by Bernouli's equation? Identify and obtain the general solution of the differential equation $2\frac{dy}{dx}-\frac{y}{x}=\frac{y^2}{x^2}$ **2+5**

2. Explain exact differential equation. Find the general solution of the differential equation $y(xy+2x^2y^2)dx+x(xy-x^2y^2)dy=0$. **2+4**

3. Write down the condition for a differential equation to be non linear and non homogeneous. **2**

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Class test: 03, Course Code: **MATH-1241**,

Course title: Ordinary, Partial Differential Equations and Coordinate Geometry

xecute the general solution of the differential equation $y + px = x^4p^2$, where $p = \frac{dy}{dx}$. **05**

sses the general solution of the following differential equations (any two including i): **05+05**

$(D^2 + 4D + 4)y = e^{3x} - e^{-2x}$

$(D^3 - D^2 - 6D)y = x^2 + 1$

$(D^2 + 3D + 2)y = e^{2x} sinx.$

CT-03

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Bangladesh Army University of Engineering & Technology CT-04

Class test: 04, Course Code: **MATH-1241**,

Course title: Ordinary, Partial Differential Equations and Coordinate Geometry

1. What do you mean by partial differential equation? What are the applications of partial differential equation in computer science & engineering? **2+4**

2. Find the partial differential equation of all spheres of radius a, having centre on the yz plane. **4**

3. Which method do you prefer to solve a linear partial differential equation of order one? Using your chosen method solve the partial differential equation $y^2 p - xyq = x(z-2y)$. **5**

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